Arunabha Majumder

Prototype Engineer | Wearable Robot Hardware, Test Rigs, 3D Printing and Assembly

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SUMMARY

I build robot hardware that people wear, and the rigs to test it. For three years I have turned my own CAD files into working shoulder exoskeleton units: printing the parts, putting in the motor, encoder and sensors, fixing what failed, and running it with real users in trials. I like a messy bench that produces a working part by Friday. SolidWorks (CSWA), 3D printing, Arduino, MATLAB and Python.

KEY RESULTS

- **12** human trial participants run on hardware I built and kept working
- **>25%** less energy from the spring and motor exoskeleton I designed and built (Best Paper, 2026)
- **<10%** force error on a test setup I built, checked against a force sensor

EXPERIENCE

Research Engineer, Wearable Robot Prototypes (PhD project VIEXO) | 2023 - Present
Aalborg University, Aalborg, Denmark

- Kept a human-worn shoulder exoskeleton running through trials with 12 participants, 3 assist modes and 2 loads by building the units from my own CAD files and BOMs, and repairing and refitting them between sessions.
- Went through several versions of each part in weeks rather than months by 3D printing them in house, fitting them on the rig the same day and sending what I learned straight back into the CAD.
- Cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026) with a spring plus 6 Nm motor actuator that I designed in SolidWorks, then built and assembled myself.
- Got force readings within 10% RMS of a reference sensor by building a test setup around a 5-bar parallel robot with variable-stiffness actuators and calibrating it against a force sensor.
- Wired motors, encoders, force sensors and DAQ into test rigs and an Arduino-based setup for functional, calibration and durability checks, and helped run lab demos. Supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Actuators (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic artificial muscles to an existing lab design, tested them on the lab's setup (valves, pressure sensors, DAQ), and used the data with the lab's neural network to set the PID gains, which brought position error down to 0.3-0.78%.

Bachelor's Capstone and Student Builds | 2017 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Designed, machined, assembled and user-tested an upper-limb rehabilitation exoskeleton. Best Major Project Award and an INR 100,000 TATA Technologies grant.
- Built a Go-Kart from scratch (2017) and a human-electric hybrid vehicle for Efficycle (2018) in the college workshop.

SKILLS

- **Workshop:** 3D printing (Bambu Lab, Ultimaker Cura), hand tools, machining and assembly (Bachelor's), test rigs and jigs
- **CAD:** SolidWorks (CSWA): reading and making parts, fixtures, assemblies and BOMs
- **Electronics and MCUs:** Arduino, motors, encoders, force/torque and pressure sensors, DAQ, pneumatic valves
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic) for test scripts and data

- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (in progress) | 2023 - Present
Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Also IEEE CONECCT 2022 (pneumatic muscle control).

REFERENCES

Available upon request.